



PS -1 Seminar
on

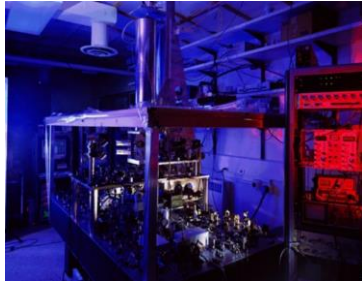
Atomic Clock

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Background

- The performances of numerous systems directly depend on the time or frequency reference they use.
- Most used technology = Quartz crystal oscillators – has its limit – accuracy - affected by external factors over time
- New technology = Atomic clocks -> most accurate time and frequency standards -> Used for international time distribution service or in global navigation satellite systems.
- Drawbacks – Huge size, High cost, Immense Power consumption
- Newest technology - **Objective:** To develop a portable, battery-powerable, stable and cheap time reference : **Ultra small atomic clock**

Background – Existing Atomic Clocks



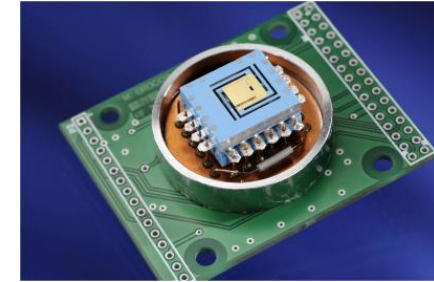
Cs Primary
standard



Commercial
Cs beam clock



Compact
atomic clock



Miniature
Atomic clock

Accuracy: 10-15
Timing error: 10ns / year
Size: 107 cm³
Power: kW
Cost: >1M\$
Frequency stability: 10-14

10-13
10μs / year
104 cm³
100's W
50k\$
Few 10-12

10-11
0.1μs / day
102 cm³
1 W
2k\$
10-11

10-10
1μs / day
10 cm³
150 mW
300\$
10-10

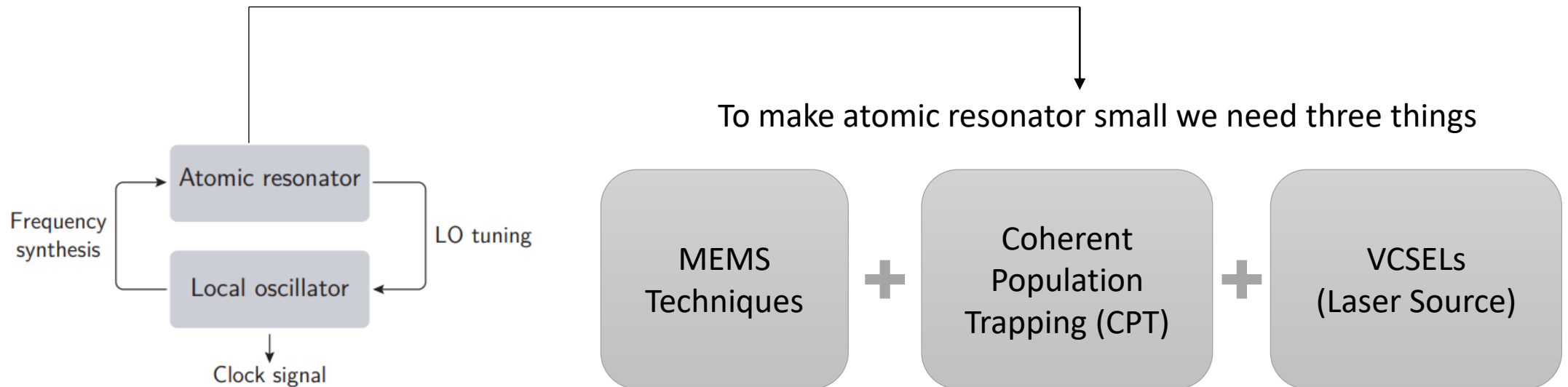
Motivation

Atomic clock has various application such as

- **Global Navigation Satellite Systems (GNSS):** Enhance accuracy and reliability in positioning, especially in environments with weak or obstructed satellite signals.
- **Telecommunications:** Improve synchronization in networks, boosting data integrity, reducing latency, and enhancing performance, crucial for 5G networks and IoT.
- **Military and Defense**
- **Research:** Enable high-precision synchronization in geophysics, astronomy, and high-energy physics experiments, facilitating accurate measurements and observations.

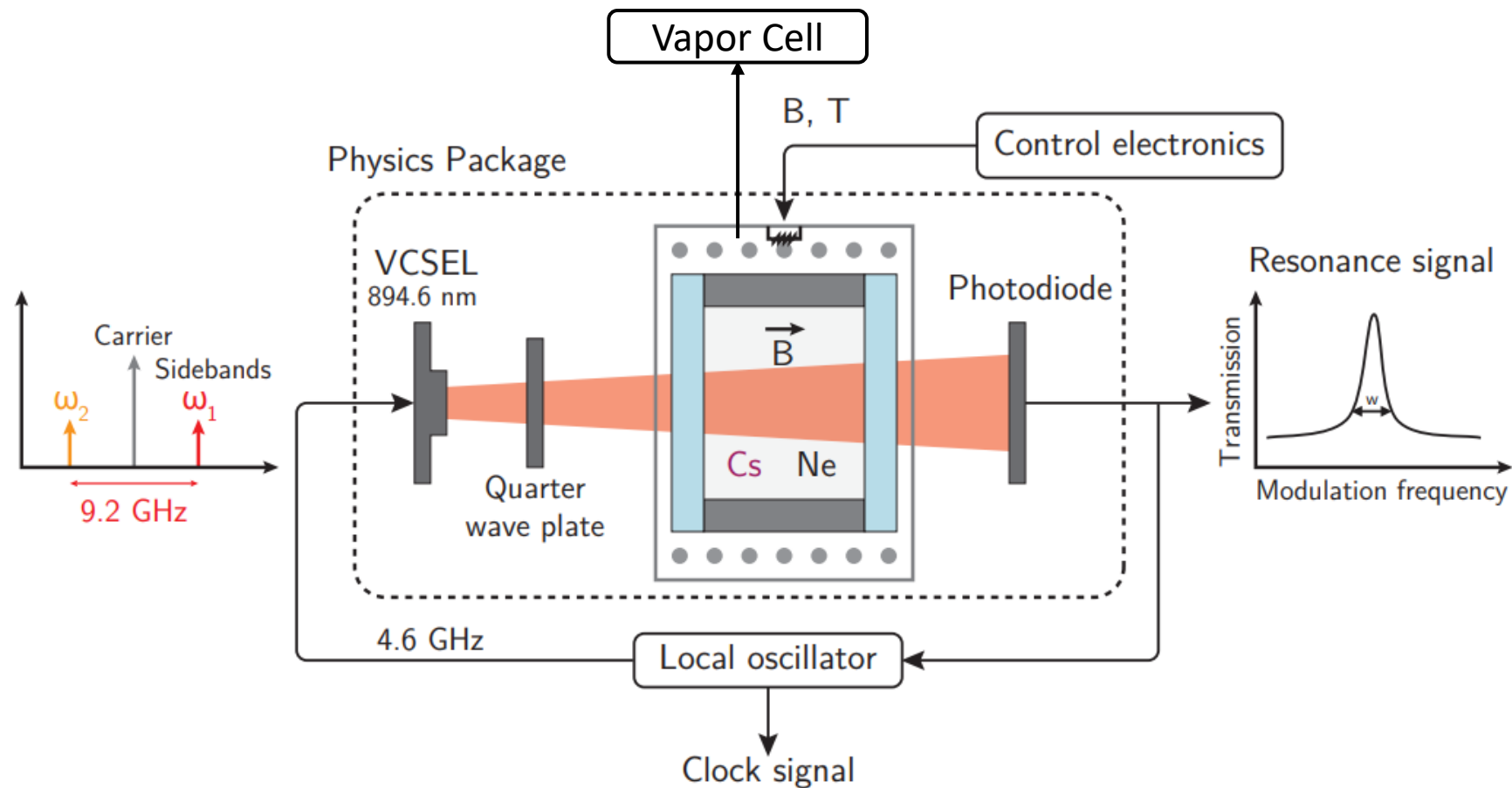
Principle

- Atomic clock: Lock the frequency of a local oscillator onto an atomic transition frequency – Positive Feedback Loop



Objective

To develop a portable, battery-powerable, stable and cheap time reference : **Ultra small atomic clock**



Our Work

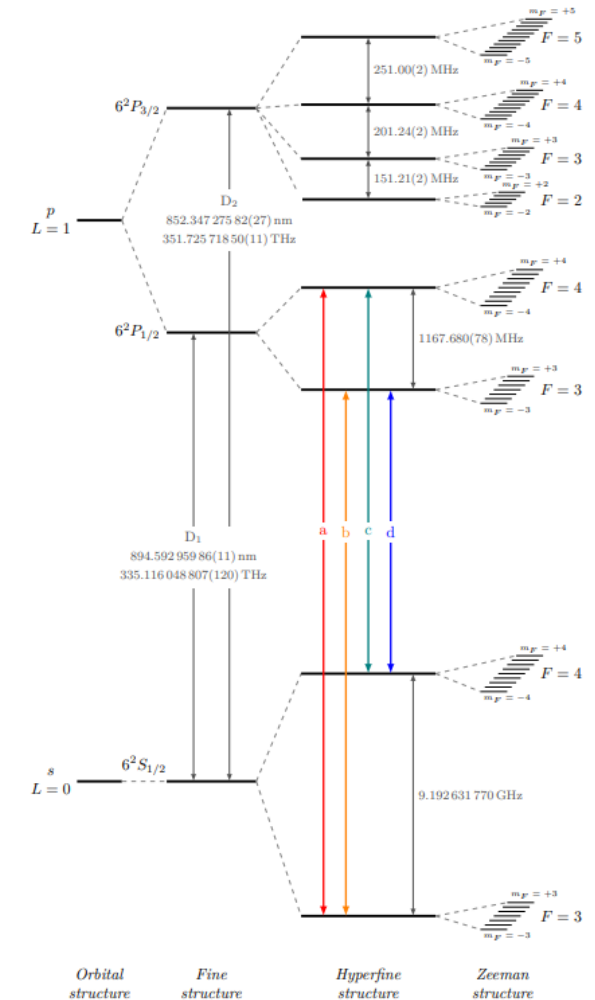
Created a full fledged GUI application to simulate performance and stability of the Atomic Clock with various parameters, including various different settings - along with Adit

Done under Mr Poorvendu – Scientist 'F' DRDO

Use different papers to collect data and functions, verify experimental results with the simulations, Do theoretical study of CPT and other physics involved in the atomic clock

GUI Application Features:

- Plot the Absorption cross section and Transmittance of the various atomic transitions under different conditions.
- Select line, buffer gas, temperature, pressure and other parameters and plot our graph.
- Calculate FWHM, Area, Peaks, Absolute/Relative frequency. Use different GUI setting to under the phenomenon better
- Plot the CPT signal with its FWHM and Amplitude under various conditions
- Plot different dependent parameters based on independent variables

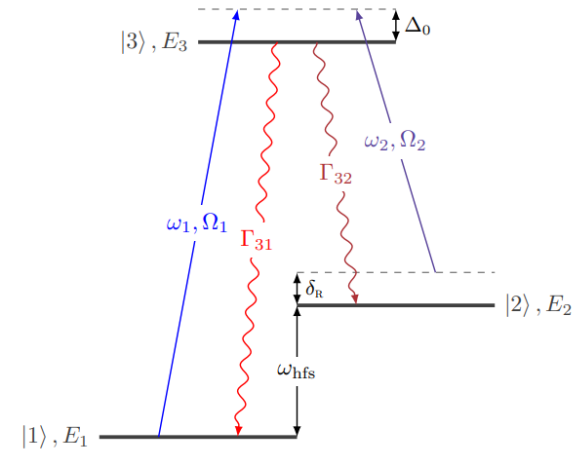


Theoretical Study

CPT Principle

An atom with 3 energy levels interacts with 2 optical fields with frequencies ω_1 and ω_2 .

- A coherent superposition of the 2 ground states occurs, leading to a transparency when $(\omega_1 - \omega_2)$ equals the clock transition frequency.
- This is used to lock a local oscillator to the atomic resonance.



Liouville Equation

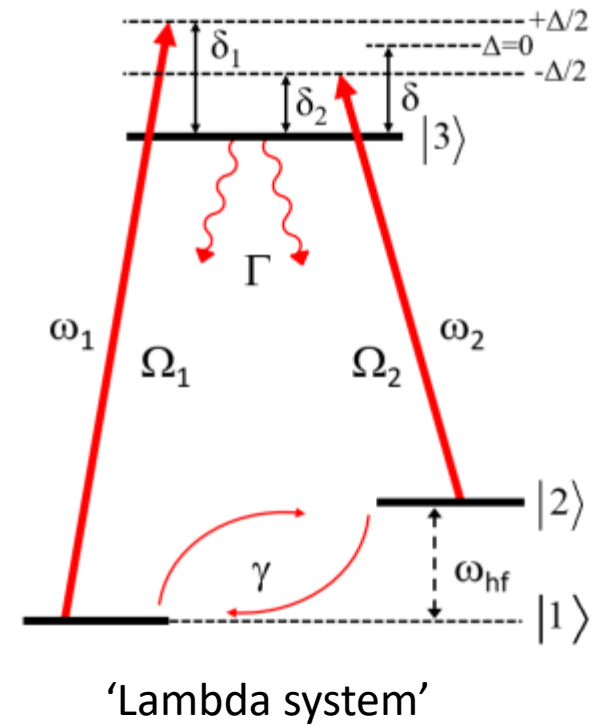
$$\frac{\partial \rho}{\partial t} = -\frac{i}{\hbar} [H\rho - \rho H^\dagger] + \dot{\rho}_{src}$$

Hamiltonian for the lambda system

$$H = \hbar \begin{bmatrix} \frac{\Delta}{2} - \frac{i\gamma}{2} & 0 & -\frac{\Omega_1}{2} \\ 0 & -\frac{\Delta}{2} - \frac{i\gamma}{2} & -\frac{\Omega_2}{2} \\ -\frac{\Omega_1}{2} & -\frac{\Omega_2}{2} & -\delta - \frac{i\Gamma}{2} \end{bmatrix}$$

Source Matrix

$$\dot{\rho}_{src} = \begin{bmatrix} \frac{\Gamma}{2}\rho_{33} + \gamma\rho_{22} & 0 & 0 \\ 0 & \frac{\Gamma}{2}\rho_{33} + \gamma\rho_{11} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



Equations obtained by solving the Liouville equation

$$\dot{\rho}_{11} = -\left(i\frac{\Omega_1}{2} \rho_{13} + \text{c.c.}\right) + \frac{\Gamma}{2} \rho_{33} - \gamma (\rho_{11} - \rho_{22})$$

$$\dot{\rho}_{13} = -i\frac{\Omega_1}{2} (\rho_{11} - \rho_{33}) - i\frac{\Omega_2}{2} \rho_{12} - \left(\frac{\Gamma}{2} + \frac{\gamma}{2} + i\left(\delta + \frac{\Delta}{2}\right)\right) \rho_{13}$$

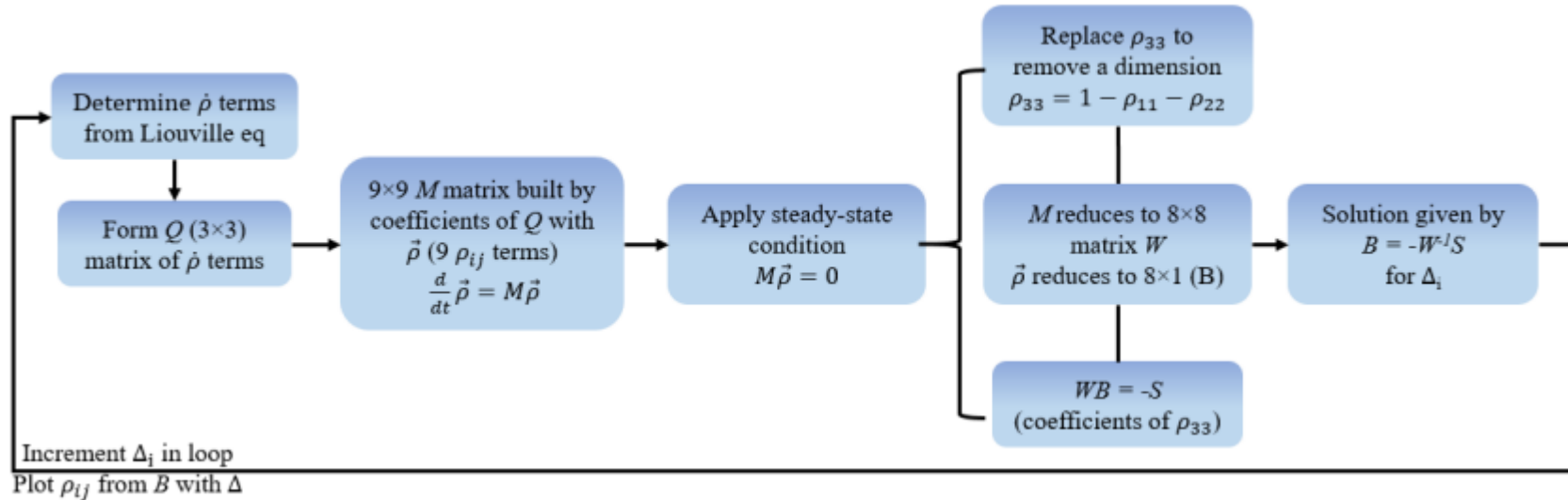
$$\dot{\rho}_{22} = -\left(i\frac{\Omega_2}{2} \rho_{23} + \text{c.c.}\right) + \frac{\Gamma}{2} \rho_{33} + \gamma (\rho_{11} - \rho_{22})$$

$$\dot{\rho}_{23} = -i\frac{\Omega_2}{2} (\rho_{22} - \rho_{33}) - i\frac{\Omega_1}{2} \rho_{21} - \left(\frac{\Gamma}{2} + \frac{\gamma}{2} + i\left(\delta - \frac{\Delta}{2}\right)\right) \rho_{23}$$

$$\dot{\rho}_{33} = \left(i\frac{\Omega_1}{2} \rho_{13} + \text{c.c.}\right) + \left(i\frac{\Omega_2}{2} \rho_{23} + \text{c.c.}\right) - \Gamma \rho_{33}$$

$$\dot{\rho}_{12} = i\frac{\Omega_1}{2} \rho_{32} - i\frac{\Omega_2}{2} \rho_{13} - (\gamma + i\Delta) \rho_{12}$$

Algorithm to solve the density matrix elements(population) in steady states



Application Screenshots

Parameters

Cs

D1

No Buffer Gas

No Buffer Gas

Buffer Gas Ratio

1

P (Torr)

100

T (C)

25

Length (mm)

1.5

Radius (mm)

10

Intensity (mW)

100

Range(GHz)

1e9

Index

33

Settings

☐

FWHM

☐

Area

☐

Detuning

☐

Peaks

☐

Legend

☐

Sweep

☐

Cross Section

☐

Fine

☐

Log

☐

Grid

☐

Ruler

☐

Dark Theme

Reset

Abs Cross-Section

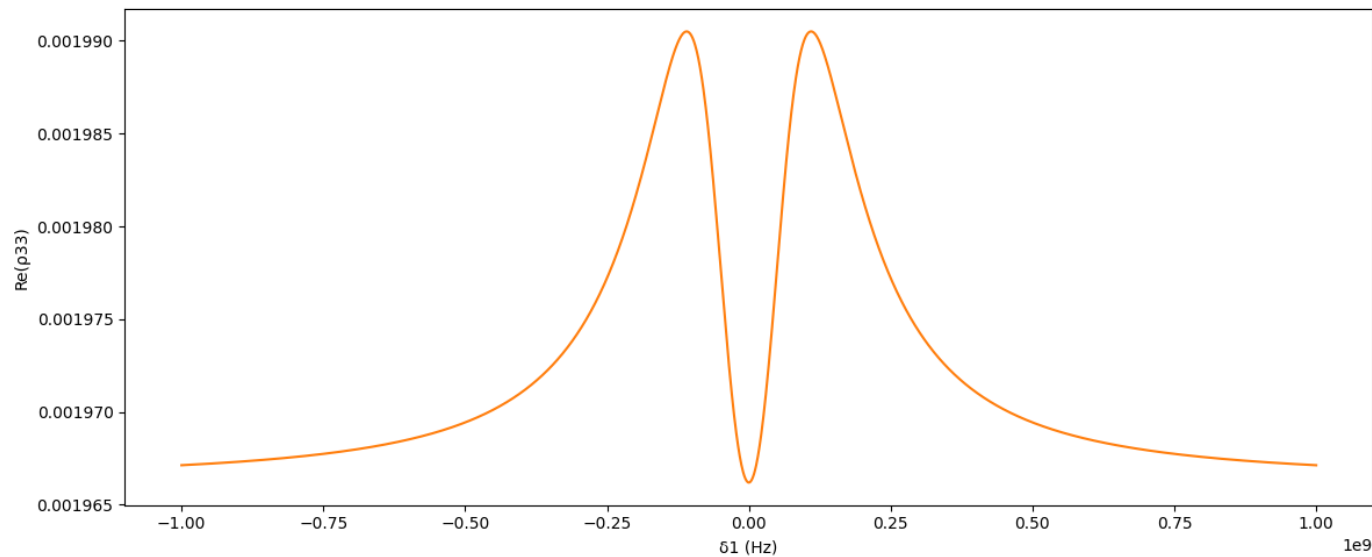
Transmittance

CPT Signal

Data

Plot parameters

CPT Signal



(x, y) = (1.46e+08, 0.00197474)

Plot CPT Warren

Plot all

Export Data

Clear Last

Clear All

Import Data

Select Transition

F

3

F'

4

mf

0

mf'

0

delta

0

62

0

Console: Now plotting...



Parameters

Rb

D1

No Buffer Gas

No Buffer Gas

Buffer Gas Ratio

1

P (Torr)

100

T (C)

25

Length (mm)

1.5

Radius (mm)

10

Intensity (mW)

100

Range(GHz)

8

Index

33

Settings

☐ FWHM☐ Area☐ Detuning☐ Peaks☐ Legend☐ Sweep☐ Cross Section☐ Fine☐ Log☐ Grid☐ Ruler☒ Dark Theme

Reset

Abs Cross-Section

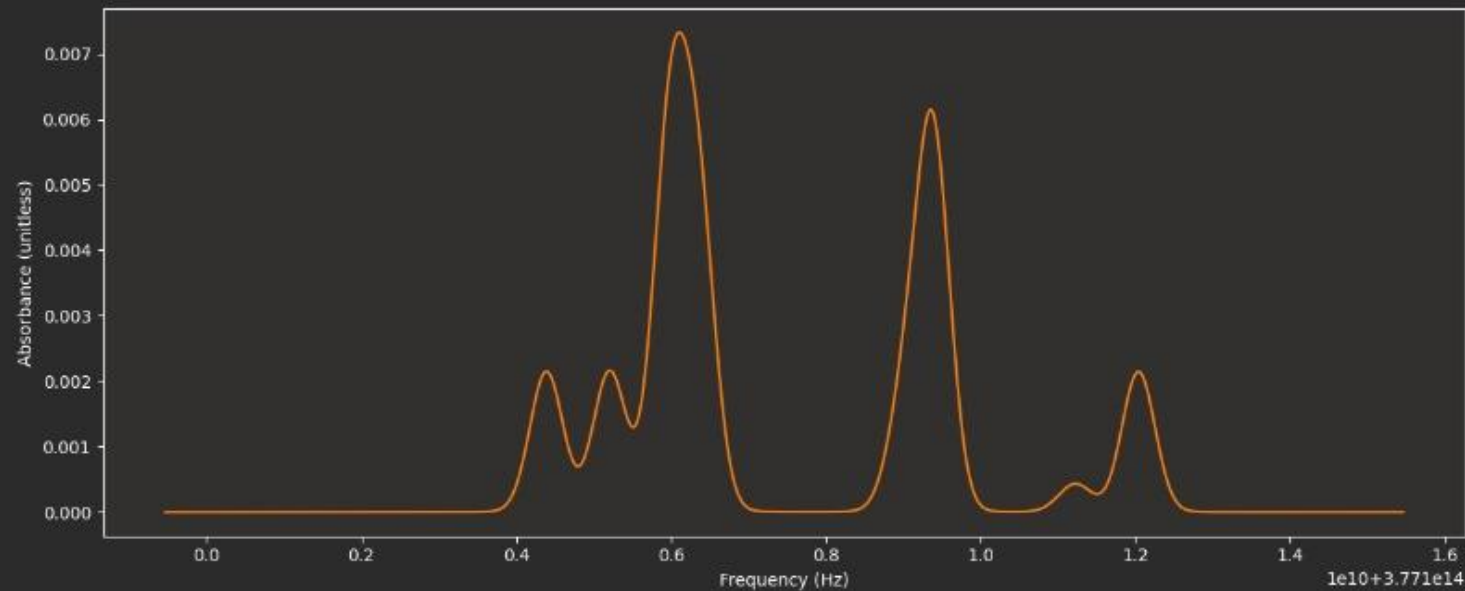
Transmittance

CPT Signal

Data

Plot parameters

Absorption Spectra



Plot Total

Plot Decomposition

Export Data

Clear Last

Clear All

Import Data

Absorption

Aff

Voigt

ΓL (Collisional)

ΓG (Doppler)

Shift

Vapor Pressure

Transmittance

Zeeman

$$\sigma(\nu) = \pi r_e c f \sum_{F, F'} A_{F, F'} V(\nu - \nu_{F, F'})$$



Console: Now plotting...

Parameters

Cs

D1

No Buffer Gas

No Buffer Gas

Buffer Gas Ratio

1

P (Torr)

100

T (C)

25

Length (mm)

1.5

Radius (mm)

10

Intensity (mW)

100

Range(GHz)

8

Index

33

Settings



FWHM



Area



Detuning



Peaks



Legend



Sweep



Cross Section



Fine



Log



Grid



Ruler

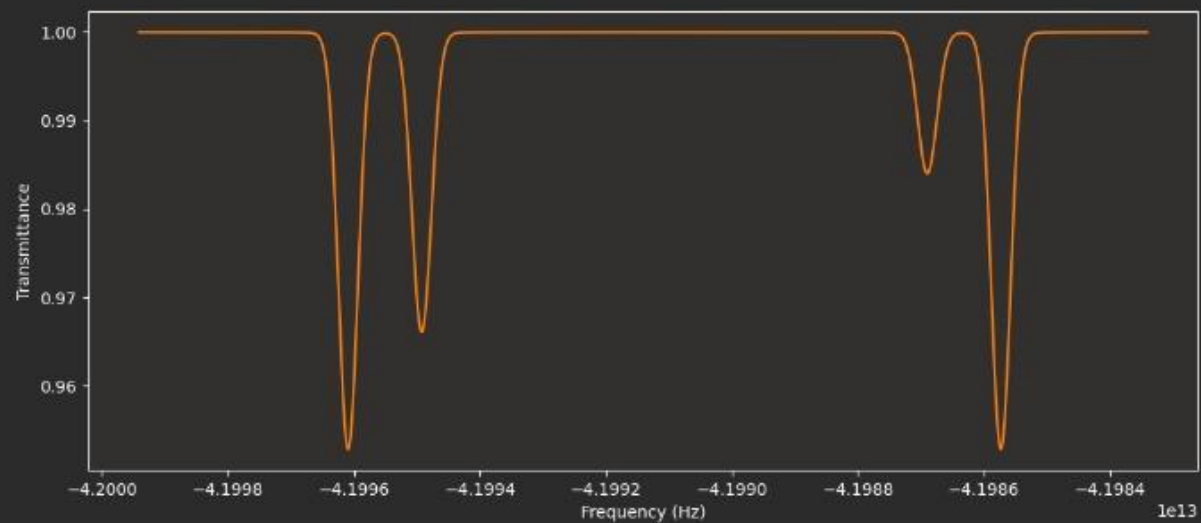


Dark Theme

Reset

Abs Cross-Section Transmittance CPT Signal Data Plot parameters

Transmittance



Plot Transmittance

Export Data

Import Data

Clear Last

Clear All

Absorption

Aff

Voigt

 Γ L (Collisional) Γ G (Doppler)

Shift

Vapor Pressure

Transmittance

Zeeman

$$\sigma(\nu) = \pi r_e c f \sum_{F,F'} A_{F,F'} V(\nu - \nu_{F,F'})$$



Console: Now plotting...

Parameters

Cs

D1

Ne

No Buffer Gas

Buffer Gas Ratio

1

P (Torr)

100

T (C)

25

Length (mm)

1.5

Radius (mm)

10

Intensity (mW)

100

Range(GHz)

8

Index

33

Settings



FWHM



Area



Detuning



Peaks



Legend



Sweep



Cross Section



Fine



Log



Grid



Ruler



Dark Theme

Reset

Abs Cross-Section

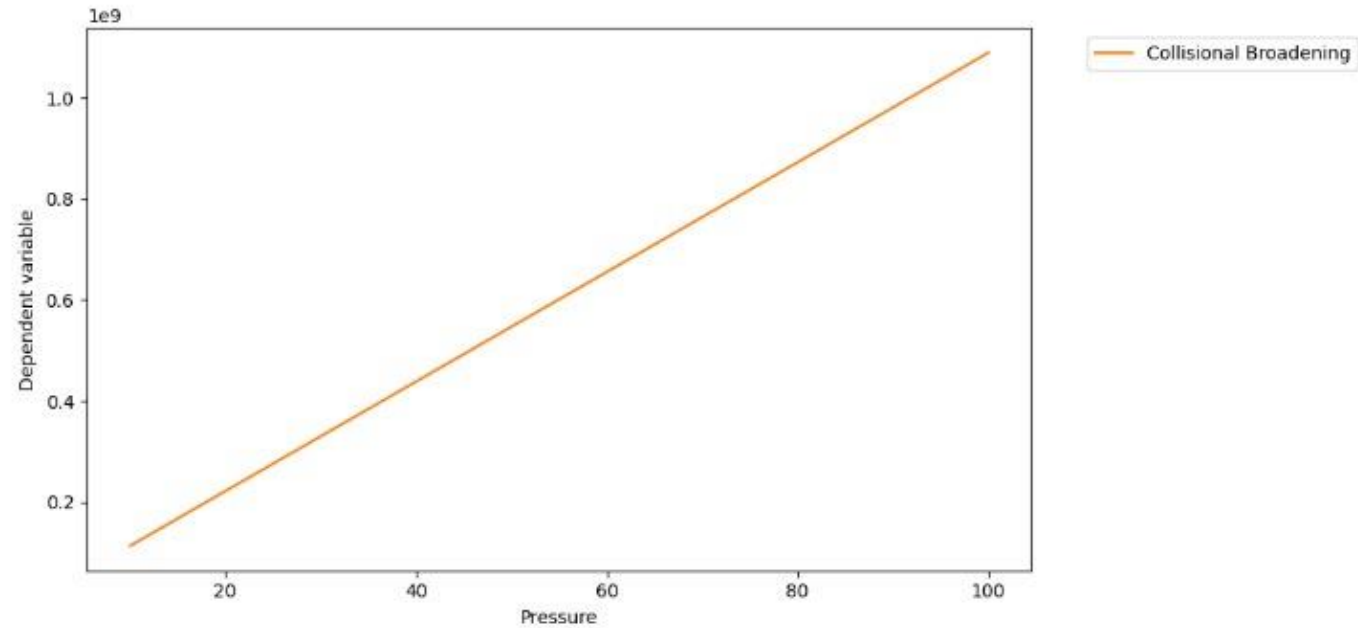
Transmittance

CPT Signal

Data

Plot parameters

Plot parameters



Plot Variation

Export Data

Import Data

Clear Last

Clear All

Pressure

Collisional Broadening

Start

10

Stop

100

Relation:

$$\Gamma_L = \gamma P + \Gamma_{\text{tot}}$$

Console: Now plotting...

Parameters

Cs

D1

No Buffer Gas

No Buffer Gas

Buffer Gas Ratio

1

P (Torr)

100

T (C)

25

Length (mm)

1.5

Radius (mm)

10

Intensity (mW)

100

Range(GHz)

8

Index

33

Settings



FWHM



Area



Detuning



Peaks



Legend



Sweep



Cross Section



Fine



Log



Grid



Ruler



Dark Theme

Reset

Abs Cross-Section

Transmittance

CPT Signal

Data

Plot parameters

View Data

v0 vg ve D0 yB delS yN M_vap M_bg R_vap R_bg I s_se s_bg f

Element

Transition

Frequency (Hz)

Cs

D1

335116048807000.0

Cs

D2

351725718500000.0

Rb

D1

377107463500000.0

Rb

D2

384230484468500.0

Edit Data

```
v0 = {'Cs': {'D1': 335116048807000.0, 'D2': 351725718500000.0}, 'Rb': {'D1': 377107463500000.0, 'D2': 384230484468500.0}}
```

```
vg = {'Cs': {'D1': [0, 0, 0, 5170855370.625, -4021776399.375, 0, 0], 'D2': [0, 0, 0, 5170855370.625, -4021776399.375, 0, 0]}, 'Rb': {'D1': [0, 4271676318.1519, -2563005979.08911, 0, 0, 0, 0], 'D2': [0, 4271676318.1519, -2563005979.08911, 0, 0, 0, 0]}, 85: {'D1': [0, 0, 1770843922.8, -1264888516.3, 0, 0, 0], 'D2': [0, 0, 1770843922.8, -1264888516.3, 0, 0, 0]}}
```

```
ve = {'Cs': {'D1': [0, 0, 0, -656820000.0, 510860000.0, 0, 0], 'D2': [0, 0, -339712800.0, -188488500.0, 12798510.0, 263890600.0, 0]}, 'Rb': {'D1': [0, -510410000.0, 306246000.0, 0, 0, 0, 0], 'D2': [-302073800.0, -229851800.0, -72911300.0, 193740800.0, 0, 0, 0]}, 85: {'D1': [0, 0, -210923000.0, 150659000.0, 0, 0, 0], 'D2': [0, -113208000.0, -83835000.0, -20435000.0, 100205000.0, 0, 0]}}
```

```
D0 = {'Cs': {'No Buffer Gas': 1, 'Ne': 1.53e-05, 'Ar': 1.34e-05, 'N2': 8.7e-06, '3-He': 0, 'He': 0, 'Kr': 0}, 'Rb': {'No Buffer Gas': 1, 'Ne': 1.53e-05, 'Ar': 1.34e-05, 'N2': 8.7e-06, '3-He': 0, 'He': 0, 'Kr': 0}}
```

```
yB = {'Cs': {'D1': {'No Buffer Gas': 0, 'Ne': 10850000.0, 'Ar': 18310000.0, 'N2': 16360000.0, '3-He': 26000000.0, 'He': 0, 'Kr': 17820000.0}, 'D2': {'No Buffer Gas': 0, 'Ne': 9810000.0, 'Ar': 16470000.0, 'N2': 19180000.0, '3-He': 22350000.0, 'He': 0, 'Kr': 15540000.0}}, 'Rb': {'D1': {'No Buffer Gas': 0, 'Ne': 9840000.0, 'Ar': 18100000.0, 'N2': 16300000.0, '3-He': 0, 'He': 20800000.0, 'Kr': 17100000.0}, 'D2': {'No Buffer Gas': 0, 'Ne': 9470000.0, 'Ar': 17700000.0, 'N2': 18300000.0, '3-He': 0, 'He': 20300000.0, 'Kr': 17200000.0}}}
```

```
delS = {'Cs': {'D1': {'No Buffer Gas': 0, 'Ne': -1600000.0, 'Ar': -6470000.0, 'N2': -7710000.0, '3-He': 6010000.0, 'He': 0, 'Kr': -5460000.0}, 'D2': {'No Buffer Gas': 0, 'Ne': -2580000.0, 'Ar': -6180000.0, 'N2': -6200000.0, '3-He': 600000.0, 'He': 0, 'Kr': -6090000.0}}, 'Rb': {'D1': {'No Buffer Gas': 0, 'Ne': -900000.0, 'Ar': -6770000.0, 'N2': -7410000.0, '3-He': 0, 'He': 5800000.0, 'Kr': -5120000.0}, 'D2': {'No Buffer Gas': 0, 'Ne': -2440000.0, 'Ar': -5760000.0, 'N2': -5790000.0, '3-He': 0, 'He': 390000.0, 'Kr': 5500000.0}}}
```

```
yN = {'Cs': {'D1': 4560000.0, 'D2': 5220000.0}, 'Rb': {'D1': 5746000.0, 'D2': 6065000.0}}
```

```
M_vap = {'Cs': 2.20894657e-25, 'Rb': {87: 1.4431606e-25, 85: 1.40999344065e-25}}
```

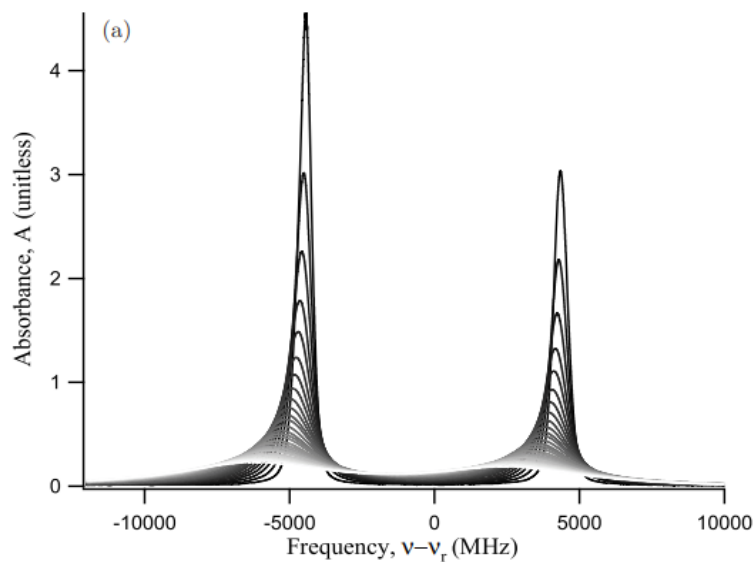
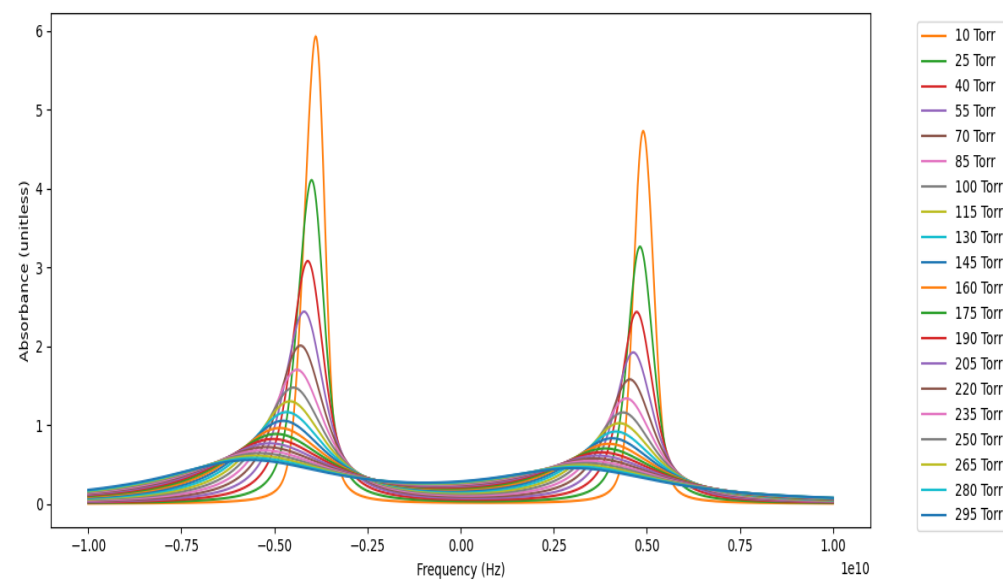
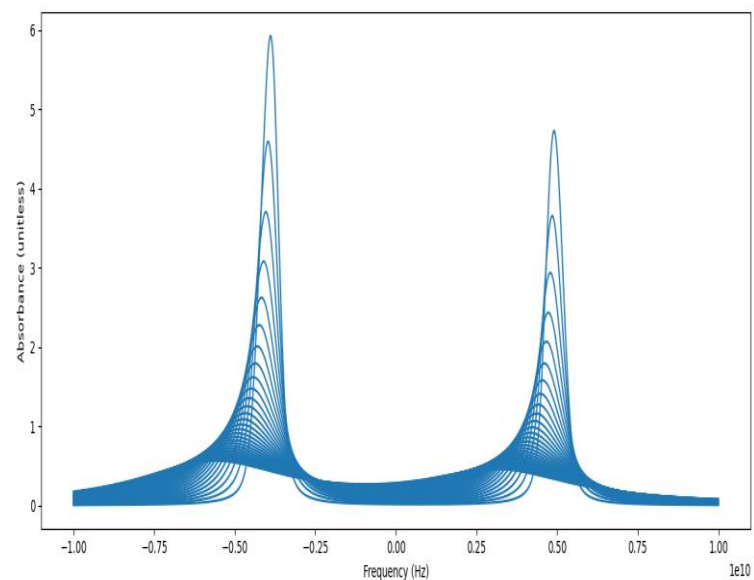
```
M_bg = {'No Buffer Gas': 1, 'Ne': 3.3509e-26, 'Ar': 6.6335e-26, 'N2': 4.6517302e-26, '3-He': 0, 'He': 0, 'Kr': 1.3914984e-25}
```

Save Changes

Reset to Default

Console: Parameters validated

Data Calibration



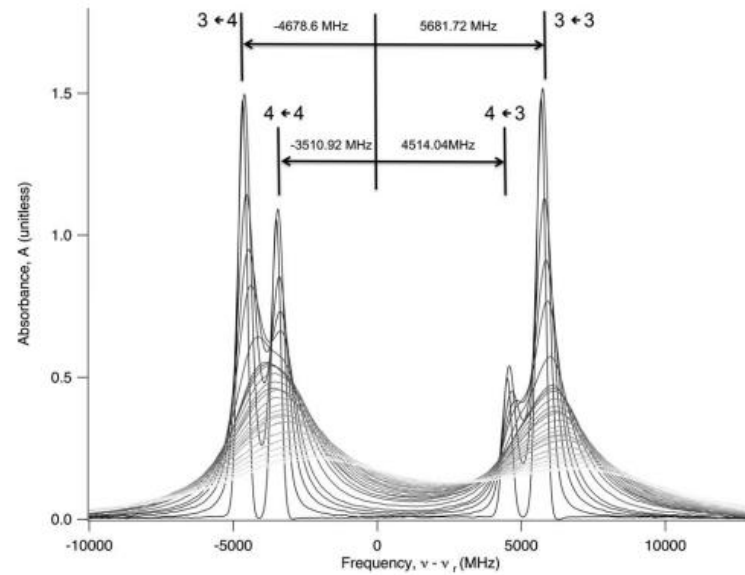
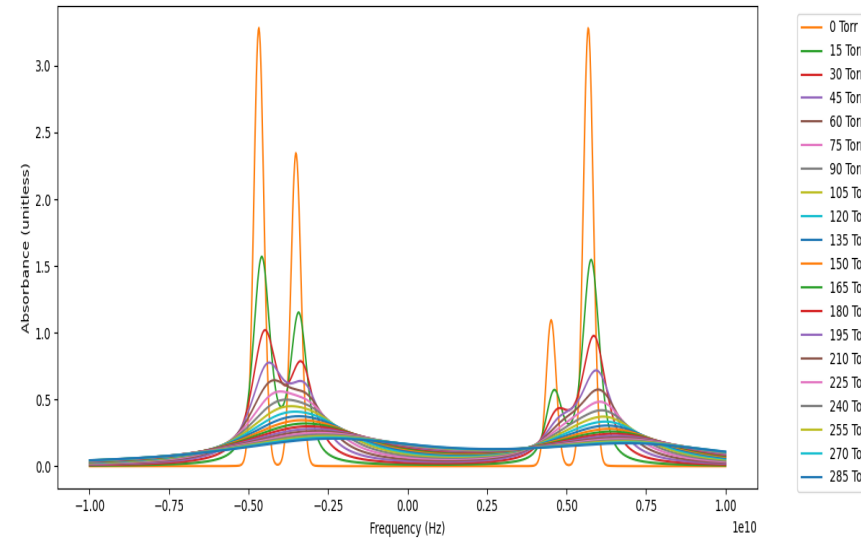
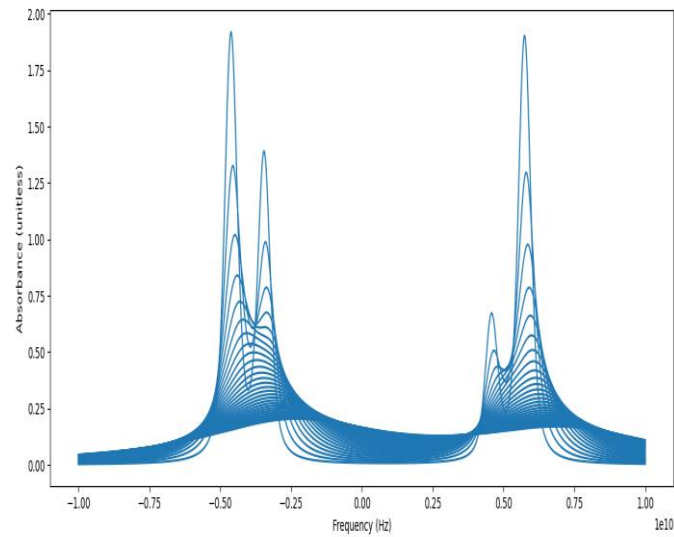
Reference

Pressure broadening and shift of the cesium D2 transition by the noble gases and N₂, H₂, HD, D₂, CH₄, C₂H₆, CF₄, and 3He with comparison to the D1 transition (2010)

Greg A. Pitz, Charles D. Fox, and Glen P. Perram*

Remarks

“The absorbance is also defined as $A = \sigma l N$ where σ is the absorption cross section, $l = 2.5$ cm is the cell path length, and $n = 1.7 \times 10^{11}$ cm⁻³ is the Cs number density at 313 K”



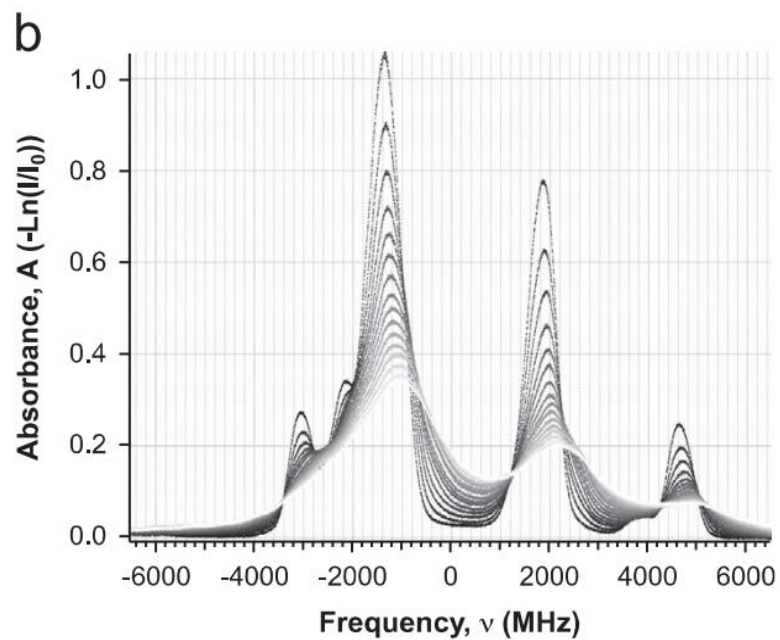
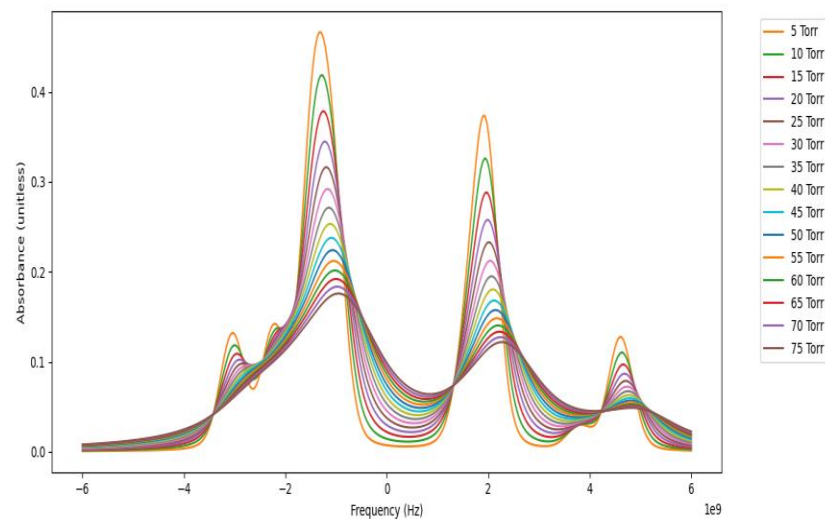
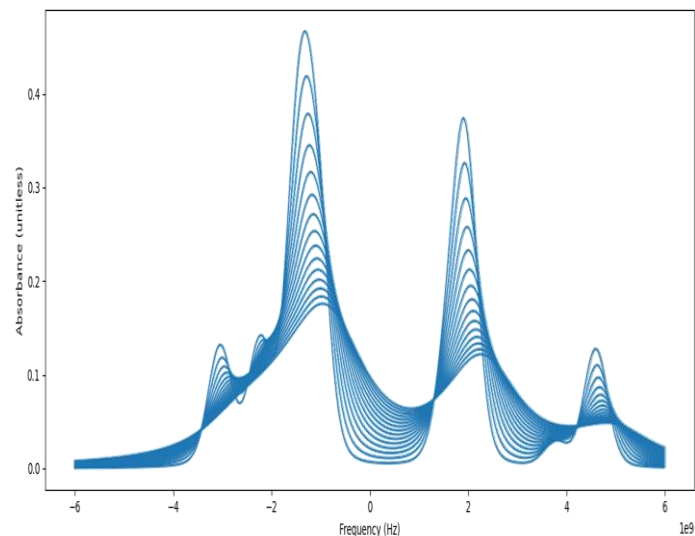
Reference

Pressure broadening and shift of the cesium D1 transition by the noble gases and N₂, H₂, HD, D₂, CH₄, C₂H₆, CF₄, and 3 He (2009)

Greg A. Pitz, Douglas E. Wertheppny, and Glen P. Perram

Remarks

“The cesium D1 hyperfine line-shape pressure broadened by 3 He from 10–300 Torr in steps of 10 Torr with the assigned $F \leftarrow F$ hyperfine transitions, where r is line center for the D1 transition.”



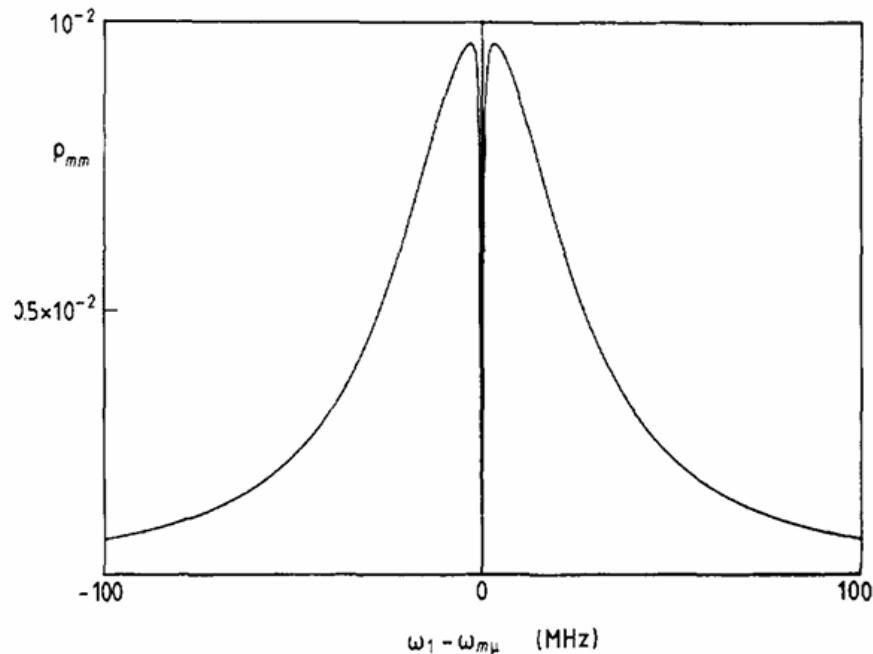
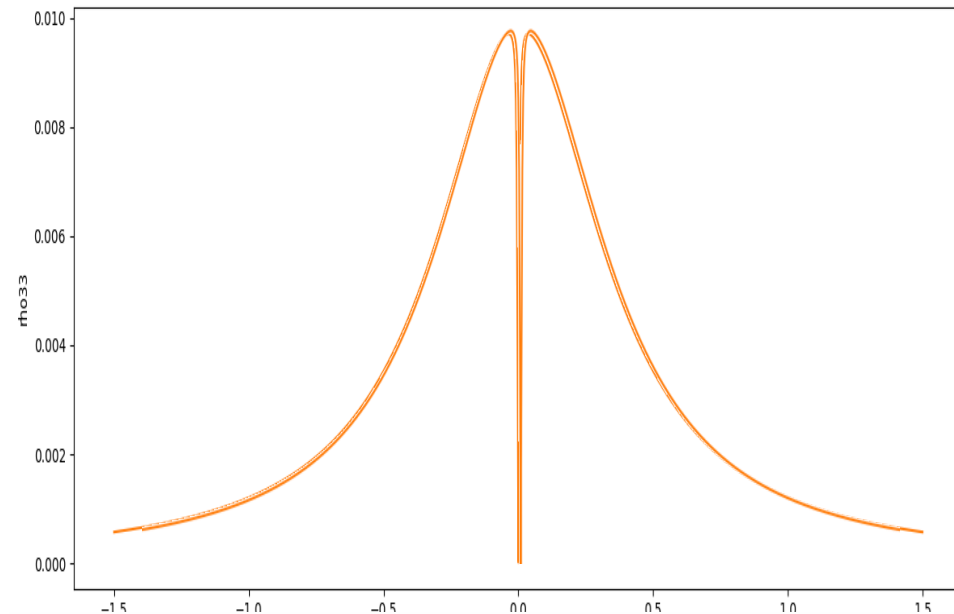
Pressure broadening and shift of the rubidium D_1 transition and potassium D_2 transitions by various gases with comparison to other alkali rates

Greg A. Pitz ^{a,*}, Andrew J. Sandoval ^a, Tiffany B. Tafoya ^a,
Wade L. Klennert ^b, David A. Hostutler ^a

^a Air Force Research Laboratory, 3550 Aberdeen Ave SE, Kirtland Air Force Base, NM 87117, USA

^b The Boeing Company, 4411 The 25 Way NE #350, Albuquerque, NM 87109, USA

An example of the resulting spectral profiles for the K D_2 and Rb D_1 broadened by helium are shown in Fig. 2(a) and (b), respectively. The hyperfine spectrum for the Rb D_1 profile is mostly resolved at low pressures. On the other



The Quantum Physics of Atomic Frequency Standards Volume 1

Jacques Vanier

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Paris Sud, Orsay

Figure 4.2.11 Illustration of optical pumping with two monochromatic lasers in a three-level system. The parameters are typical of caesium atoms in a cell and are as follows: $\gamma_1 \approx \gamma_2 \approx 2 \times 10^5 \Gamma$; $\Gamma = 0.37 \times 10^8 \text{ s}^{-1}$. We assume that the intensity of the lasers is weak with $\omega_{1m\mu} = \omega_{1m\mu'} = \frac{1}{10} \Gamma$. One laser is kept resonant with one transition while the frequency of the other laser is swept through resonance. The abscissa, 0, corresponds to resonance of both lasers with the respective transitions (adapted from Orriols 1979).

Literature Survey

- **Vincent Maurice. Design, microfabrication and characterization of alkali vapor cells for miniature atomic frequency references. Atomic Physics [physics.atom-ph]. Université de Franche-Comté, 2016**
- R. Boudot, P. Dziuban, M. Hasegawa, R. K. Chutani, S. Galliou, V. Giordano, C. Gorecki; Coherent population trapping resonances in Cs–Ne vapor microcells for miniature clocks applications. J. Appl. Phys. 1 January 2011; 109 (1): 014912.
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- Jia S, Jiang Z, Jiao B, Liu X, Pan Y, Song Z, Qu J. The Microfabricated Alkali Vapor Cell with High Hermeticity for Chip-Scale Atomic Clock. Applied Sciences. 2022; 12(1):436.
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THANK YOU!